State Space Methods for Flight Vehicles

AE 695

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Definition of State of System

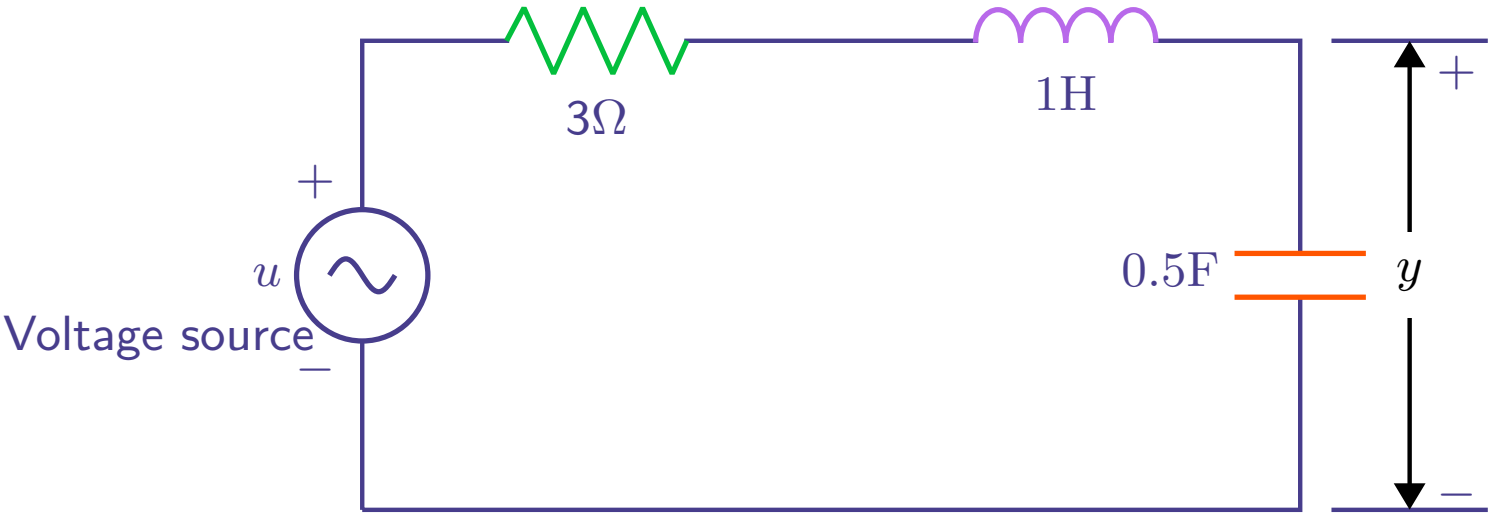
- Input output description applicable only for initially relaxed systems.
- **Output depends not only on input but also on initial conditions**.

State

The state of the system at time t_0 is the amount of information at t_0 that, together with $\boldsymbol{u}_{[t_0,\infty)}$, determines uniquely the behavior of the system for all $t \geq t_0$.

- **Behavior of system**: All responses including system state
- State at any time **summarizes the essential information** about past inputs required for the **prediction of future behavior**.
- Different inputs might lead to the same state at a given time, but their future behavior would be the same for the same future inputs.





- With known initial current and voltage across the capacitor, the behavior of the network is uniquely determinable for any input.
- Inductor current and capacitor voltage as the state of the system.
- Transfer function from u to y for this network system

$$g(s) = \frac{y(s)}{u(s)} = \frac{2}{(s+1)(s+2)} = \frac{2}{s+1} - \frac{2}{s+2}$$

- **Impulse response**: $g(t) = 2e^{-t} - 2e^{-2t}$
- With system **relaxed at** t_0, $\forall\, t \geq t_0$

$$y(t) = \int_{t_0}^{t} g(t-\tau)u(\tau)d\tau$$

- With the system not relaxed at t_0, the system output

$$y(t) = \int_{-\infty}^{t_0} g(t-\tau)u(\tau)d\tau + \int_{t_0}^{t} g(t-\tau)u(\tau)d\tau$$

- Effect of input $u_{(-\infty,t_0)}$ on the system output $y_{[t_0,\infty)}$

$$\int_{-\infty}^{t_0} g(t-\tau)u(\tau)d\tau = 2e^{-t}\underbrace{\int_{-\infty}^{t_0} e^{\tau}u(\tau)d\tau}_{c_1} - 2e^{-2t}\underbrace{\int_{-\infty}^{t_0} e^{2\tau}u(\tau)d\tau}_{c_2}$$

- System output at a time instant $t = t_0$

$$\boxed{y(t_0) = 2c_1e^{-t_0} - 2c_2e^{-2t_0}}$$

- Taking the time derivative of the output

$$\dot{y}(t) = -2c_1e^{-t} + 4c_2e^{-2t} + g(0)u(t) + \int_{t_0}^{t} \frac{\partial}{\partial t}g(t-\tau)u(\tau)d\tau$$

$$\boxed{\frac{d}{dt}\int_{t_0}^{t} g(t-\tau)u(\tau)d\tau = g(t-\tau)u(\tau)|_{t=\tau} + \int_{t_0}^{t} \frac{\partial}{\partial t}g(t-\tau)u(\tau)d\tau}$$



- With $g(0) = 0$, $\boxed{\dot{y}(t_0) = -2c_1 e^{-t_0} + 4c_2 e^{-2t_0}}$
- Solving for the coefficients c_1 and c_2

$$\boxed{c_1 = 0.5e^{t_0}[2y(t_0) + \dot{y}(t_0)], \quad c_2 = 0.5e^{2t_0}[y(t_0) + \dot{y}(t_0)]}$$

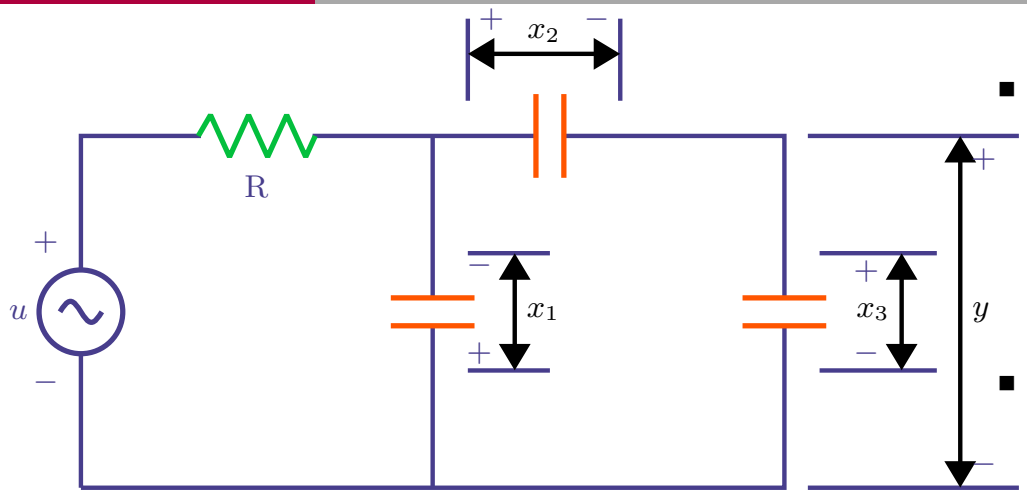
- With system not relaxed at $t = t_0$, the system output $\forall\, t \geq t_0$

$$\boxed{y(t) = [2y(t_0) + \dot{y}(t_0)]e^{-(t-t_0)} - [y(t_0) + \dot{y}(t_0)]e^{-2(t-t_0)} + \int_{t_0}^{t} g(t-\tau)u(\tau)d\tau}$$

- **Information about $y(t_0)$ and $\dot{y}(t_0)$ required to uniquely determine $y(t)$**
- Relaxedness of system at t_0 is **not necessary** to determine system behavior
- **Quantities $(y(t_0), \dot{y}(t_0))$ or (c_1, c_2) qualify for state of system**



Systems and State Vectors



- With all capacitor voltages known, network behavior is uniquely determinable for any input, thus a set of capacitor voltages as state
- Using Kirchoff's voltage law, $x_1 + x_2 + x_3 = 0$

- Only two capacitor voltages sufficient to uniquely determine system behavior
- Any two capacitor voltages qualify as the state

- **Unit-time delay**: $y(t) = u(t-1) \, \forall \, t$
- To uniquely determine $y_{[t_0,\infty)}$ from $u_{[t_0,\infty)}$, we need $u_{[t_0-1,t_0)}$
- Thus, $u_{[t_0-1,t_0)}$ as state of unit-time delay system

State choice not unique, may not be a physical quantity, or may be finite or infinite in numbers.

Systems and State Vectors

- **State vector**: Finite-dimensional column vector that includes all information (variables) necessary for determining system behavior with known future inputs.
- **State variables**: Components of the state vector
- **State space** $\sum$: A linear space in which the state vector ranges.

Dynamical equations

The set of equations that describes the unique relations between the input, output, and state of the system.

State and output equations:

$$\dot{\boldsymbol{x}} = \boldsymbol{h}\left(\boldsymbol{x}(t), \boldsymbol{u}(t), t\right), \quad \boldsymbol{y} = \boldsymbol{g}\left(\boldsymbol{x}(t), \boldsymbol{u}(t), t\right)$$

Lumped and Distributed Systems

Lumped: Number of state variables is finite or its state is a finite vector

Distributed: Infinitely many state variables



Linearity of Dynamical Systems

- Input-state-output pair: $\{\boldsymbol{u}_{[t_0,\infty)},\, \boldsymbol{x}(t_0)\} \rightarrow \{\boldsymbol{x}_{[t_0,\infty)},\, \boldsymbol{y}_{[t_0,\infty)}\}$
- An input-state-output pair is admissible if system can generate such a pair.

A system is said to be linear iff for any two admissible pairs

$$\{\boldsymbol{x}^1(t_0),\, \boldsymbol{u}^1_{[t_0,\infty)}\} \rightarrow \{\boldsymbol{x}^1_{[t_0,\infty)},\, \boldsymbol{y}^1_{[t_0,\infty)}\}$$

$$\{\boldsymbol{x}^2(t_0),\, \boldsymbol{u}^2_{[t_0,\infty)}\} \rightarrow \{\boldsymbol{x}^2_{[t_0,\infty)},\, \boldsymbol{y}^2_{[t_0,\infty)}\}$$

and any real numbers α_1 and α_2, the following pair

$$\{\alpha_1\boldsymbol{x}^1(t_0) + \alpha_2\boldsymbol{x}^2(t_0), \alpha_1\boldsymbol{u}^1_{[t_0,\infty)} + \alpha_2\boldsymbol{u}^2_{[t_0,\infty)}\}$$
$$\rightarrow \{\alpha_1\boldsymbol{x}^1_{[t_0,\infty)} + \alpha_2\boldsymbol{x}^2_{[t_0,\infty)}, \alpha_1\boldsymbol{y}^1_{[t_0,\infty)} + \alpha_2\boldsymbol{y}^2_{[t_0,\infty)}\}$$

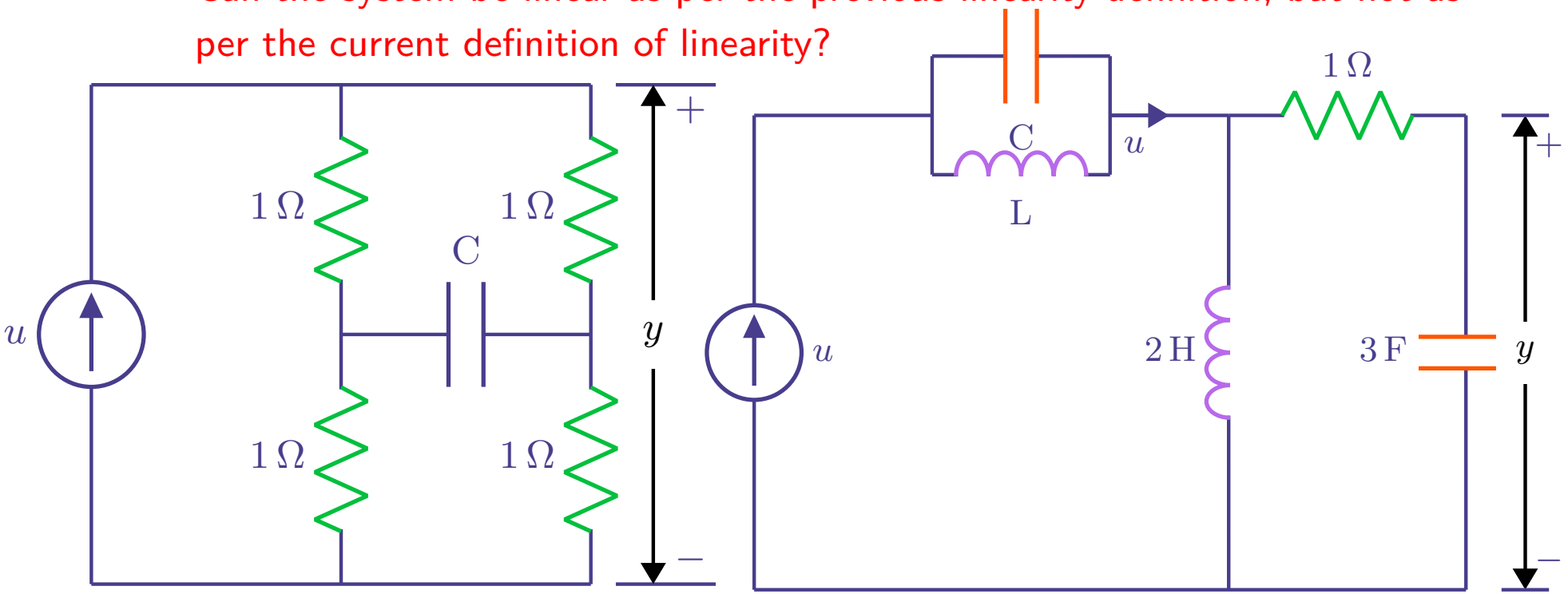
is also admissible. Otherwise, the system is said to be nonlinear.

- **Superposition**: **Additivity** ($\alpha_1 = \alpha_2 = 1$), **Homogeneity** ($\alpha_2 = 0$)
- Superposition must hold not only at output but also at all state variables
- Superposition must hold for zero and nonzero initial states



Linearity of Dynamical Systems

- Can the system be linear as per the previous linearity definition, but not as per the current definition of linearity?



- With a nonlinear capacitor, if the voltage across the capacitor is zero at $t_0 = 0$, then it will remain at zero for all time, irrespective of inputs.
- As the LC loop is in series with the current source, its behavior will not be transmitted to the output.
- Thus, these systems are linear systems with the old linearity definition, **but nonlinear as per the new definition for dynamical systems**.

Linearity of Dynamical Systems

- **Necessary condition for linear system**: $\{\mathbf{0}, \mathbf{0}\} \rightarrow \{\mathbf{0}_{[t_0,\infty)}, \mathbf{0}_{[t_0,\infty)}\}$
- With zero initial state and zero input, system responses are identically zero.
- **Response of linear system due to** $\{\boldsymbol{x}(t_0), \boldsymbol{u}_{[t_0,\infty)}\}$: Response due to $\{\boldsymbol{x}(t_0), \mathbf{0}\}$ + Response due to $\{\mathbf{0}, \boldsymbol{u}_{[t_0,\infty)}\}$
- **Zero input response**: Response due to $\{\boldsymbol{x}(t_0), \mathbf{0}\}$
- **Zero state response**: Response due to $\{\mathbf{0}, \boldsymbol{u}_{[t_0,\infty)}\}$
- For a linear dynamical system

$$\boxed{\boldsymbol{h}(\boldsymbol{x}, \boldsymbol{u}, t) = \boldsymbol{A}(t)\boldsymbol{x}(t) + \boldsymbol{B}(t)u(t), \quad \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u}, t) = \boldsymbol{C}(t)\boldsymbol{x}(t + \boldsymbol{E}(t)\boldsymbol{u}(t)}$$

$$\boxed{\dot{\boldsymbol{x}} = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{u}, t) = \boldsymbol{A}(t)\boldsymbol{x}(t + \boldsymbol{B}(t)\boldsymbol{u}(t), \quad \boldsymbol{y} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u}, t) = \boldsymbol{C}(t)\boldsymbol{x}(t + \boldsymbol{E}(t)\boldsymbol{u}(t)}$$

- **Unique solution**: Every entry of $\boldsymbol{A}(t)$ be a continuous function of t over $(-\infty, \infty)$
- Linear time-varying (LTV) dynamical system



Time Invariance of Dynamical Systems

A system is said to be **time-invariant** iff for any admissible pair

$$\{\boldsymbol{x}(t_0), \boldsymbol{u}_{[t_0,\infty)}\} \rightarrow \{\boldsymbol{x}_{[t_0,\infty)}, \boldsymbol{y}_{[t_0,\infty)}\}$$

and any real number α, the pair

$$\{Q_\alpha \boldsymbol{x}(t_0), Q_\alpha \boldsymbol{u}_{[t_0,\infty)}\} \rightarrow \{Q_\alpha \boldsymbol{x}_{[t_0,\infty)}, Q_\alpha \boldsymbol{y}_{[t_0,\infty)}\}$$

is also admissible. Otherwise, the system is said to be time-varying.

- For same initial states and waveform of excitations, waveform of responses will always be the same, no matter what instant the excitations are applied.
- For LTI systems, $\boxed{\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}(t), \quad \boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{E}\boldsymbol{u}(t)}$
- For LTI systems, responses are independent of initial time, thus $t_0 = 0$.
- **State-space**: n-dimensional real vector space $(\mathbb{R}^n, \mathbb{R})$
- $\boldsymbol{A}$: Linear operator that maps $\sum$ to itself, with basis vectors n_i
- **Dynamical systems are all causal**.



Transfer Function Matrix for State Space Systems

- For LTI systems, using the Laplace transform

$$s\boldsymbol{x}(s) - \boldsymbol{x}_0 = \boldsymbol{A}\boldsymbol{x}(s) + \boldsymbol{B}\boldsymbol{u}(s),\ \boldsymbol{y}(s) = \boldsymbol{C}\boldsymbol{x}(s) + \boldsymbol{E}\boldsymbol{u}(s)$$

$$\boldsymbol{x}(s) = (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}_0 + (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{u}(s)$$

$$\boldsymbol{y}(s) = \boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}_0 + \boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{u}(s) + \boldsymbol{E}\boldsymbol{u}(s)$$

- Determinant of $s\boldsymbol{I} - \boldsymbol{A}$ different from zero, thus its inverse always exists.
- Assuming $\boldsymbol{x}_0 = \boldsymbol{0}$ and using comparison

$$\boldsymbol{y}(s) = \underbrace{\left[\boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{E}\right]}_{\boldsymbol{G}(s)}\boldsymbol{u}(s),\ \boldsymbol{G}(s) = \frac{1}{\det(s\boldsymbol{I} - \boldsymbol{A})}\boldsymbol{C}[\text{Adj}(s\boldsymbol{I} - \boldsymbol{A})]\boldsymbol{B} + \boldsymbol{E}$$

- Why is $\boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}$ always strictly proper?
- $\boldsymbol{G}(\infty) = \boldsymbol{E}$

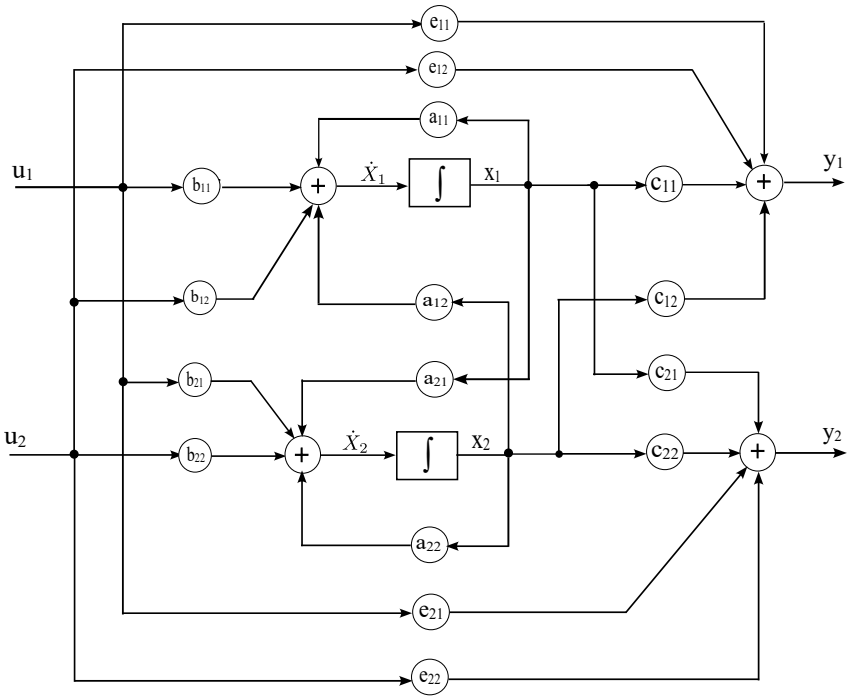


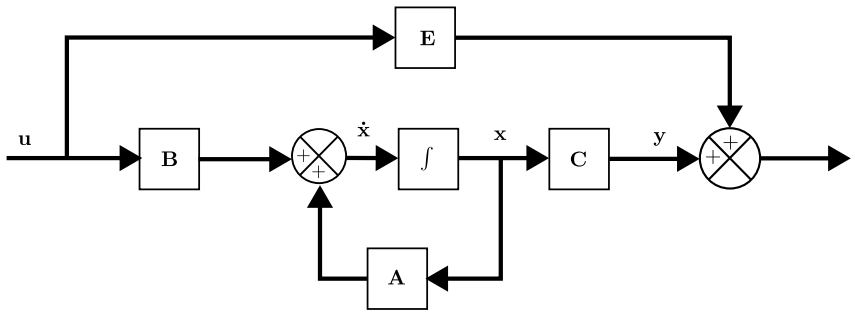
Linear Dynamical Equations

- Two-dimensional dynamical equation

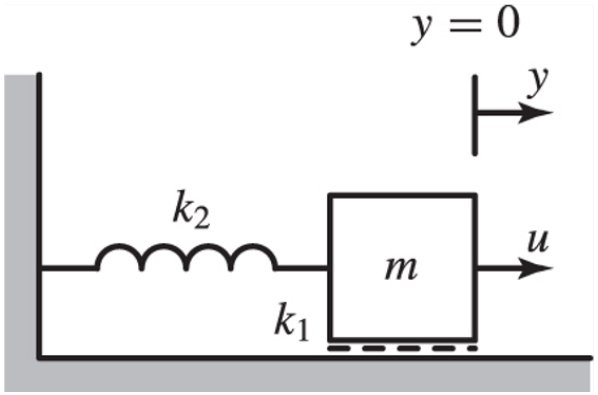
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_{\boldsymbol{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}}_{\boldsymbol{B}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

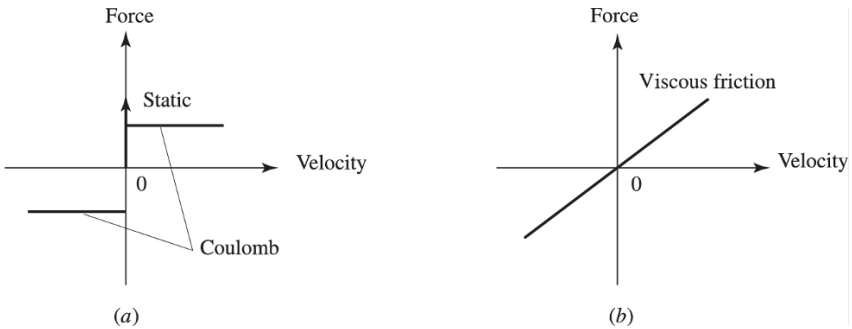
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \underbrace{\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}}_{\boldsymbol{C}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} e_{11} & e_{12} \\ e_{21} & e_{22} \end{bmatrix}}_{\boldsymbol{E}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

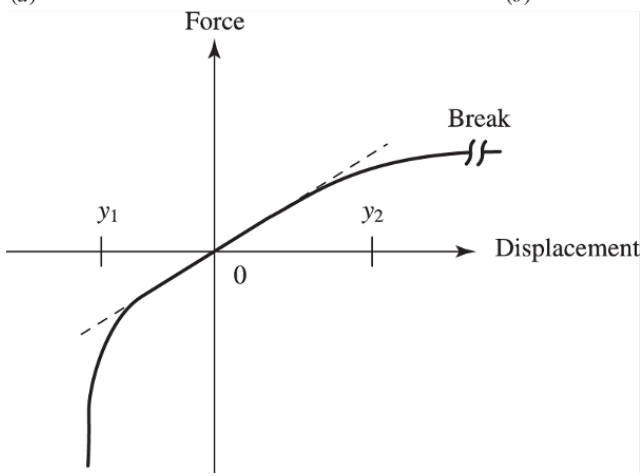




Example: Spring-Mass System







- Friction force: $f = k_1 \dfrac{dy}{dt}$
- Spring force: $k_2 y$
- Using Newton's law

$$m\frac{d^2y}{dt^2} = u - k_1\frac{dy}{dt} - k_2 y$$

- With $x_1 = y$ and $x_2 = \dot{y}$

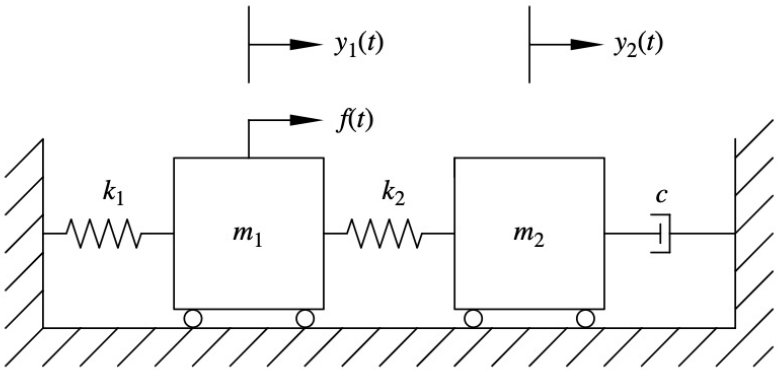
$$\dot{x}_1 = x_2, \quad m\dot{x}_2 = u - k_1 x_2 - k_2 x_1$$

- System equations

$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1 \\ -\dfrac{k_2}{m} & -\dfrac{k_1}{m}\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix} + \begin{bmatrix}0 \\ \dfrac{1}{m}\end{bmatrix} u$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix}$$

Example: Translational Mechanical System



- Using Newton's second law applied to each mass

$$m_1\ddot{y}_1 + k_1 y_1 - k_2(y_2 - y_1) = f$$

$$m_2\ddot{y}_2 + c\dot{y}_2 + k_2(y_2 - y_1) = 0$$

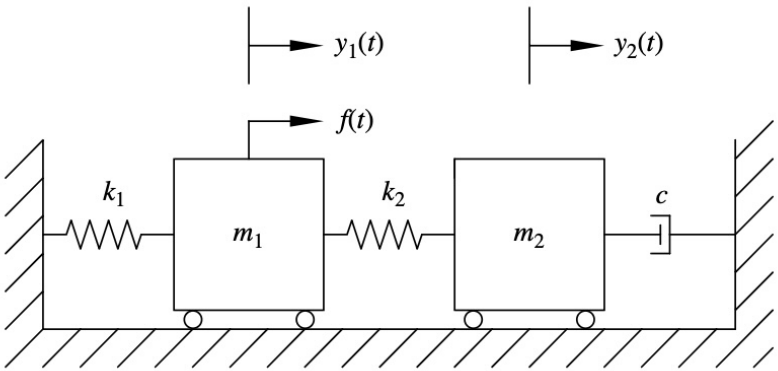
- Energy-storage elements: two springs and two masses

- State variables in terms of mass displacements and velocities
- With $x_1 = y_1, x_2 = y_2 - y_1, x_3 = \dot{y}_1$ and $x_4 = \dot{y}_2$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ -\dfrac{k_1}{m_1} & -\dfrac{k_2}{m_1} & 0 & 0 \\ 0 & -\dfrac{k_2}{m_2} & 0 & -\dfrac{c}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{m_1} \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^\top$$

Example: Translational Mechanical System



- Using Newton's second law applied to each mass

$$m_1\ddot{y}_1 + k_1 y_1 - k_2(y_2 - y_1) = f$$

$$m_2\ddot{y}_2 + c\dot{y}_2 + k_2(y_2 - y_1) = 0$$

- With $x_1 = y_1, x_2 = y_2, x_3 = \dot{y}_1$ and $x_4 = \dot{y}_2$

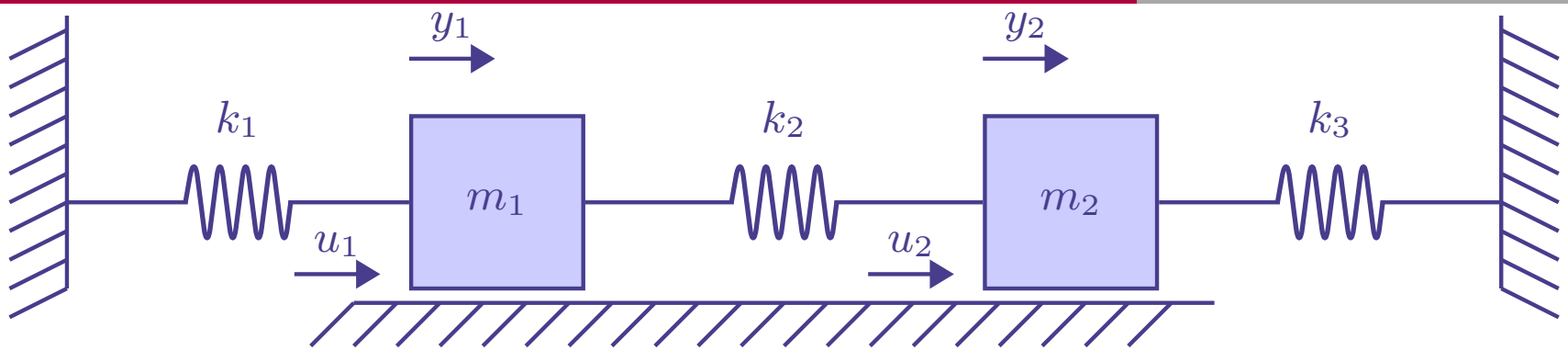
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\dfrac{k_1+k_2}{m_1} & -\dfrac{k_2}{m_1} & 0 & 0 \\ \dfrac{k_2}{m_2} & -\dfrac{k_2}{m_2} & 0 & -\dfrac{c}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{m_1} \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^\top$$

- **Different choices of state variable lead to different representations.**



Example: Translational Mechanical System



- With Newton's second law applied to each mass

$$m_1\ddot{y}_1 + (k_1 + k_2)y_1 - k_2 y_2 = u_1$$

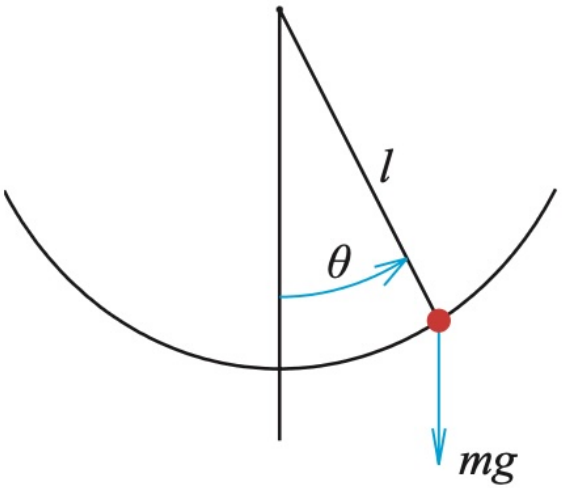
$$m_2\ddot{y}_2 - k_2 y_1 + (k_1 + k_2)y_2 = u_2$$

- State variables: $x_1 = y_1,\ x_2 = \dot{y}_2,\ x_3 = y_2,\ x_4 = \dot{y}_2$
- State equations

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\dfrac{k_1+k_2}{m_1} & 0 & \dfrac{k_2}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \dfrac{k_2}{m_2} & 0 & -\dfrac{k_1+k_2}{m_2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \dfrac{1}{m_1} & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{m_1} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^\top$$

Example: Inverted Pendulum



- Rigid rod with zero mass
- Free to swing in the vertical plane
- A frictional force resisting the motion, proportional to the speed of the bob with a coefficient of friction k
- Equation of motion in the tangential direction

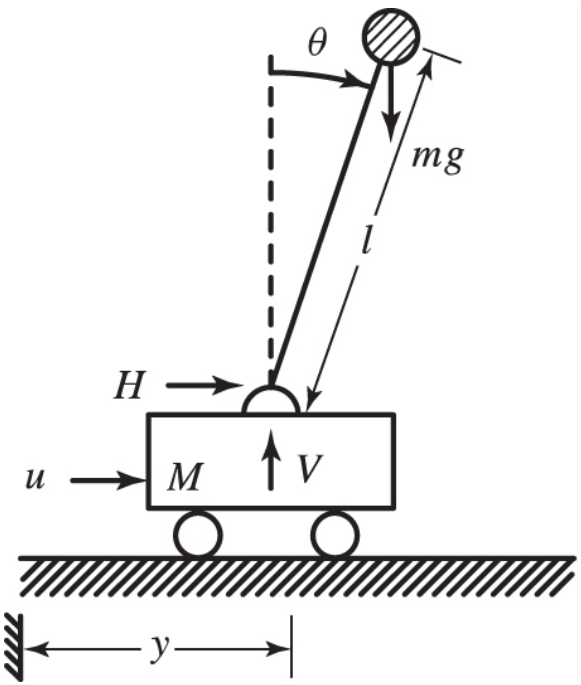
$$ml\ddot{\theta} = -mg\sin\theta - kl\dot{\theta}$$

- Equilibrium points at $\theta = 0, \pm\pi, \pm 2\pi, \ldots$
- System states: $x_1 = \theta, \quad x_2 = \dot{\theta}$
- In state-space form, the system equations

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\dfrac{g}{l}\sin x_1 - \dfrac{k}{m}x_2 \end{bmatrix}, \quad y = x_1$$



Example: Inverted Pendulum on Cart



- Movement of cart and pendulum in one plane
- Friction, mass of the stick, and wind gusts neglected
- Model for space booster at takeoff

- Using Newton's law for linear movements

$$M\frac{d^2y}{dt^2} = u - H$$

$$H = m\frac{d^2(y + l\sin\theta)}{dt^2} = m\ddot{y} + ml(\cos\theta\ddot{\theta} - \sin\theta(\dot{\theta})^2)$$

$$mg - V = m\frac{d^2(l\cos\theta)}{dt^2} = ml(-\sin\theta\ddot{\theta} - \cos\theta(\dot{\theta})^2)$$

- Using Newton's law for rotational movement around the hinge

$$ml^2\ddot{\theta} = mgl\sin\theta - m\ddot{y}l\cos\theta$$

- With small $\theta, \dot{\theta}$, and squared terms

$$\sin\theta \approx \theta, \quad \cos\theta \approx 1$$

$$M\ddot{y} = u - m\ddot{y} - ml\ddot{\theta}$$
$$ml^2\ddot{\theta} = mgl\theta - m\ddot{y}l$$

Example: Inverted Pendulum on Cart

- Governing equations for inverted-pendulum and cart motion

$$(M+m)\ddot{y} + ml\ddot{\theta} = u, \qquad l\ddot{\theta} - g\theta + \ddot{y} = 0$$

- With some algebraic manipulations

$$\ddot{y} = -\frac{mg}{M}\theta + \frac{1}{M}u, \qquad \ddot{\theta} = \frac{(M+m)}{Ml}g\theta - \frac{1}{Ml}u$$

- With system state variables

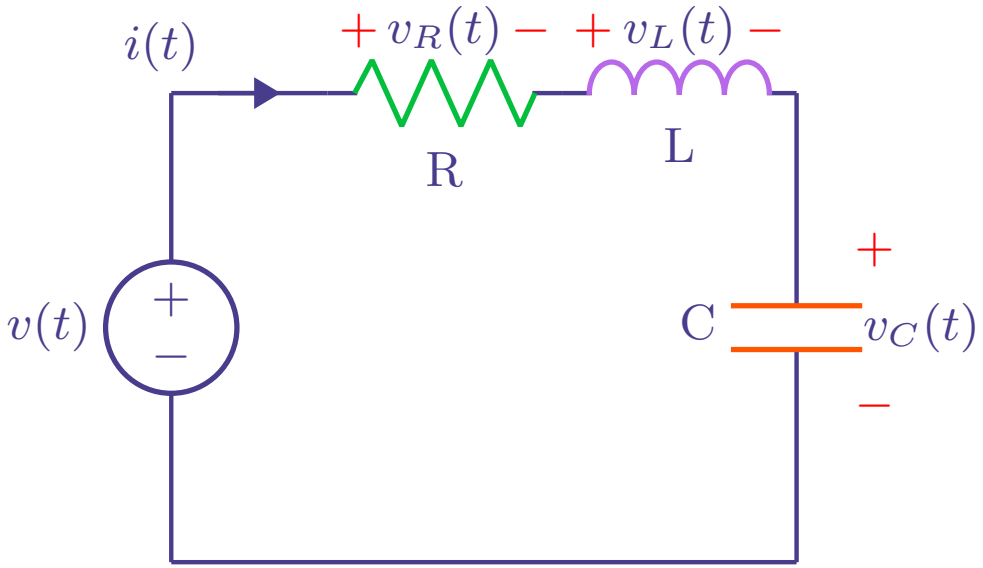
$$x_1 = y,\ x_2 = \dot{y},\ x_3 = \theta,\ x_4 = \dot{\theta}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M+mg)}{Ml} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{Ml} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^\top$$



Example: Series RLC Circuit



- State variables:
  - Current i through the inductor
  - Voltage v_C across the capacitor

$$x_1 = i, \quad x_2 = v_C$$

- Input and output: v and i

- Capacitor voltage

$$C\frac{dv_C}{dt} = i \implies \dot{x}_2 = \frac{1}{C}x_1$$

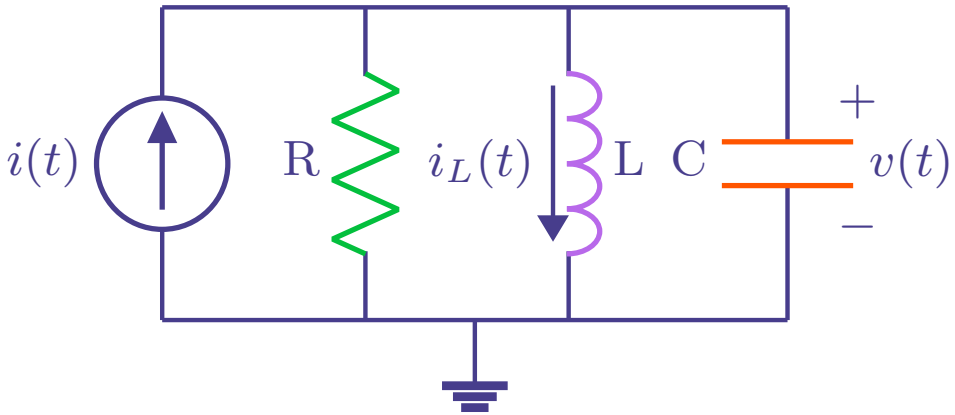
- Kirchoff's voltage law:

$$v = iR + v_C + v_L \implies L\frac{di}{dt} + v_C + iR = v \implies \dot{x}_1 = -\frac{R}{L}x_1 - \frac{1}{L}x_2 + \frac{1}{L}u$$

- State equations

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-\frac{R}{L} & -\frac{1}{L}\\ \frac{1}{C} & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}\frac{1}{L}\\ 0\end{bmatrix}u, \quad y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Example: Parallel RLC Circuit



- State variables:
  - Current i_L through the inductor
  - Voltage v across the capacitor

$$x_1 = i_L, \quad x_2 = v$$

- Input and output: i and v

- Inductor's voltage-current relationship

$$x_2 = L\frac{di_L}{dt} = L\dot{x}_1 \implies \dot{x}_1 = \frac{1}{L}x_2$$

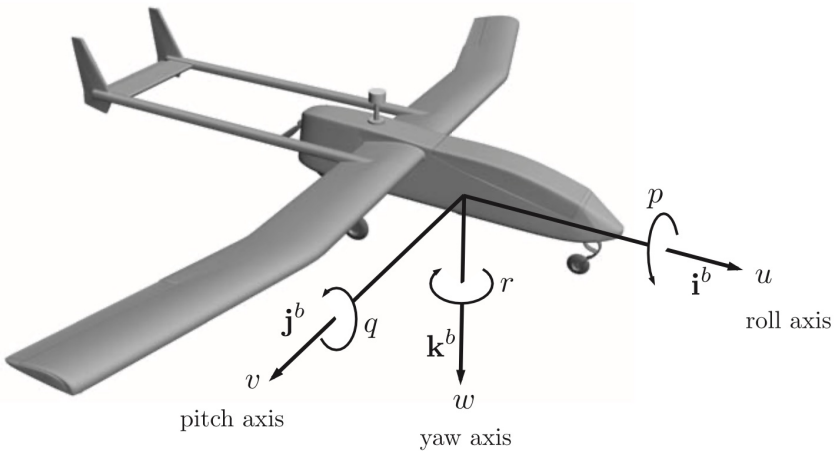
- Kirchhoff's current law:

$$\frac{v}{R} + i_L + C\frac{dv}{dt} = i \implies \dot{x}_2 = -\frac{1}{RC}x_2 - \frac{1}{C}x_1 + \frac{1}{C}u$$

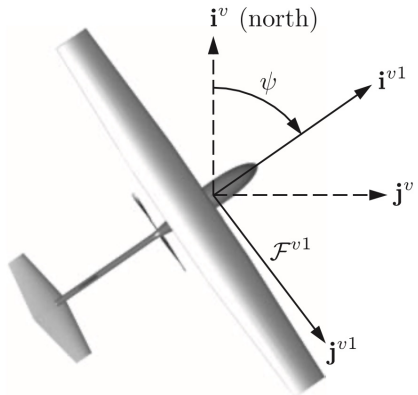
- State equations

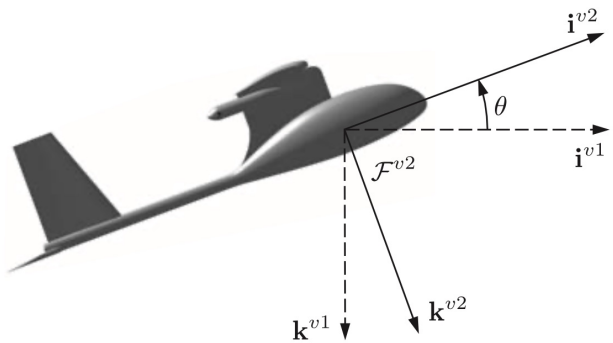
$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC}\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix} + \begin{bmatrix}0 \\ \frac{1}{C}\end{bmatrix}u, \quad y = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix}$$

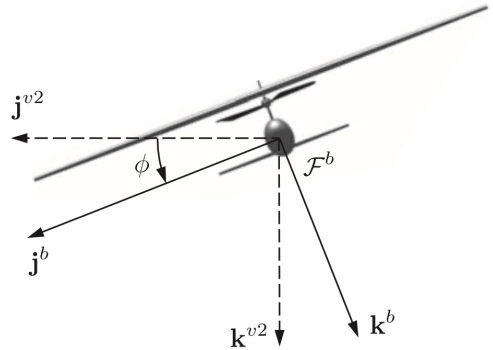
Unmanned Aerial Vehicles: Axes and Symbols



Name	*Description*
p_n	Inertial north position of the MAV along $\mathbf{i}^i$ in $\mathcal{F}^i$
p_e	Inertial east position of the MAV along $\mathbf{j}^i$ in $\mathcal{F}^i$
p_d	Inertial down position (negative of altitude) of the MAV measured along $\mathbf{k}^i$ in $\mathcal{F}^i$
u	Body frame velocity measured along $\mathbf{i}^b$ in $\mathcal{F}^b$
v	Body frame velocity measured along $\mathbf{j}^b$ in $\mathcal{F}^b$
w	Body frame velocity measured along $\mathbf{k}^b$ in $\mathcal{F}^b$
ϕ	Roll angle defined with respect to $\mathcal{F}^{v2}$
θ	Pitch angle defined with respect to $\mathcal{F}^{v1}$
ψ	Heading (yaw) angle defined with respect to $\mathcal{F}^v$
p	Roll rate measured along $\mathbf{i}^b$ in $\mathcal{F}^b$
q	Pitch rate measured along $\mathbf{j}^b$ in $\mathcal{F}^b$
r	Yaw rate measured along $\mathbf{k}^b$ in $\mathcal{F}^b$







$$\begin{pmatrix} \dot{p}_n \\ \dot{p}_e \\ \dot{p}_d \end{pmatrix} = \begin{pmatrix} \mathsf{C}_\theta \mathsf{C}_\psi & \mathsf{S}_\phi s_\theta \mathsf{C}_\psi - \mathsf{C}_\phi \mathsf{S}_\psi & \mathsf{C}_\phi \mathsf{S}_\theta \mathsf{C}_\psi + \mathsf{S}_\phi \mathsf{S}_\psi \\ \mathsf{C}_\theta \mathsf{S}_\psi & \mathsf{S}_\phi \mathsf{S}_\theta \mathsf{S}_\psi + \mathsf{C}_\phi \mathsf{C}_\psi & \mathsf{C}_\phi \mathsf{S}_\theta \mathsf{S}_\psi - \mathsf{S}_\phi \mathsf{C}_\psi \\ -\mathsf{S}_\theta & \mathsf{S}_\phi \mathsf{C}_\theta & \mathsf{C}_\phi \mathsf{C}_\theta \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

$$\begin{pmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{pmatrix} = \begin{pmatrix} rv - qw \\ pw - ru \\ qu - pv \end{pmatrix} + \frac{1}{m} \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}$$

$$\begin{pmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \Gamma_1 pq - \Gamma_2 qr \\ \Gamma_5 pr - \Gamma_6 (p^2 - r^2) \\ \Gamma_7 pq - \Gamma_1 qr \end{pmatrix} + \begin{pmatrix} \Gamma_3 & 0 & \Gamma_4 \\ 0 & 1/J_y & 0 \\ \Gamma_4 & 0 & \Gamma_8 \end{pmatrix} \begin{pmatrix} l \\ m \\ n \end{pmatrix}$$



References

References

- **Chen, Chi-Tsong, "Linear System Theory and Design", CBS College Publishing, 1984.**

